Let $\vec{\Gamma} = (\Gamma_1, .., \Gamma_n) : (\Omega, \mathcal{A}) \to (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ random vector. We say that Γ is Gaussian if $\forall \vec{l} \in \mathbb{R}^n$ $\vec{l} \cdot \vec{\Gamma}$ is Gaussian.

Teorema 1. *$\vec{\Gamma}$ is Gaussian iff $\forall \vec{\xi} \in \mathbb{R}^n$ $\mathbb{E}[exp(i\vec{\xi} \cdot \vec{\Gamma})] = exp(i\mathbb{E}[\vec{\xi} \cdot \vec{\Gamma}] - \frac{Var(\vec{\xi} \cdot \vec{\Gamma})}{2})$.*

Proof.

- $\Rightarrow$: Let $\vec{\xi} \in \mathbb{R}^n$. Since $\vec{\Gamma}$ is Gaussian, we have $\vec{\xi} \cdot \vec{\Gamma}$ Gaussian. Then $\forall \eta \in \mathbb{R} \phi_{\vec{\xi} \cdot \vec{\Gamma}}(\eta) = exp(i\mathbb{E}[\vec{\xi} \cdot \vec{\Gamma}]\eta - \frac{Var(\vec{\xi} \cdot \vec{\Gamma})\eta^2}{2})$. Taking $\eta = 1$ we obtain:
$$\mathbb{E}[exp(i\vec{\xi}\vec{\Gamma})] = exp(i\mathbb{E}[\vec{\xi} \cdot \vec{\Gamma}] - \frac{Var(\vec{\xi} \cdot \vec{\Gamma})}{2})$$

- $\Leftarrow$:

$$\begin{aligned}
\phi_{\vec{\xi} \cdot \vec{\Gamma}}(\eta) =& \mathbb{E}[exp(i\vec{\xi} \cdot \vec{\Gamma}\eta)] \\
=& \mathbb{E}[exp(i(\eta\vec{\xi}) \cdot \vec{\Gamma})] \\
=& exp(i\mathbb{E}[\eta\vec{\xi} \cdot \vec{\Gamma}] - \frac{Var(\eta\vec{\xi} \cdot \vec{\Gamma})}{2}) \\
=& exp(i\mathbb{E}[\vec{\xi} \cdot \vec{\Gamma}]\eta - \frac{Var(\vec{\xi} \cdot \vec{\Gamma})\eta^2}{2})
\end{aligned}$$

□